

Cybersecurity & Blockchain

KGSP Enrichment [Week 3 & 4 – 2026]





Mudassar Aslam

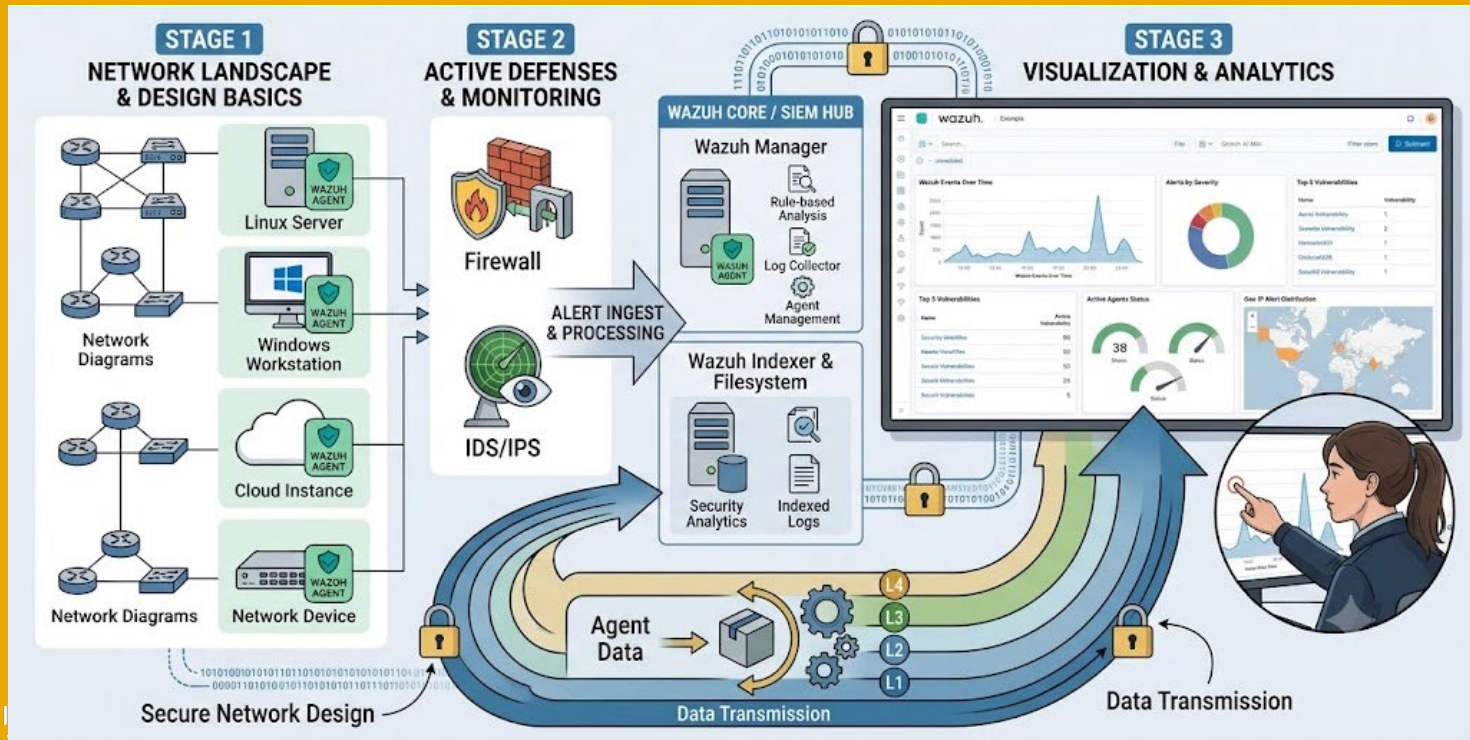
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Day 04 – Securing Networks





Basics of Networking



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Computer Networking

- Computers are powerful but become even more powerful when connected to each other
- Computer Networking **connects** them
- Networking allows: Emails, Browsing, Printing, Streaming Videos, etc.
- Every device in a Network must have an address, called **IP Address**, if it needs to communicate with another device (computer, mobile, camera, car, temperature sensor, etc.)
- There are different types of networks

Network Types



Local area networks (LANs)

- Homes & Offices
- Connects Computers, Printers, Storage



Wide area networks (WANs)

- Connects offices in different locations
- LAN to WAN -> Internet



Wi-Fi networks

- Wireless LANs



Bluetooth networks

- Personal area networks (PANs)
- Connects computers and peripherals



Near-field communication (NFC)

- Few centimeters apart



IP Address

- Internet Service -> IP Addresses
 - Telephone Service -> Telephone number
 - Postal Service -> Street address
- In most cases, IP addresses are written in what's known as the dotted quad notation of IPv4.
- A system's IP address uniquely identifies it on a network.
- The following two IP addresses are involved in every network communication:
 - Source address
 - Destination address

IPv4 Address Format (Dotted Decimal Notation)

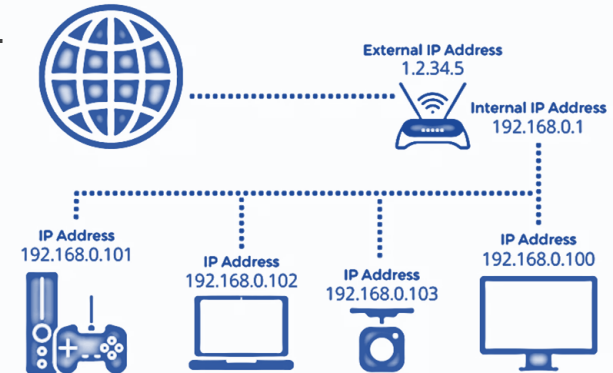
123.89.46.72

First Octet Second Octet Third Octet Fourth Octet

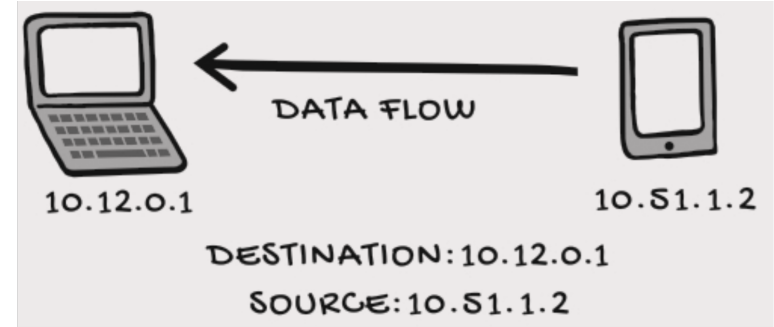
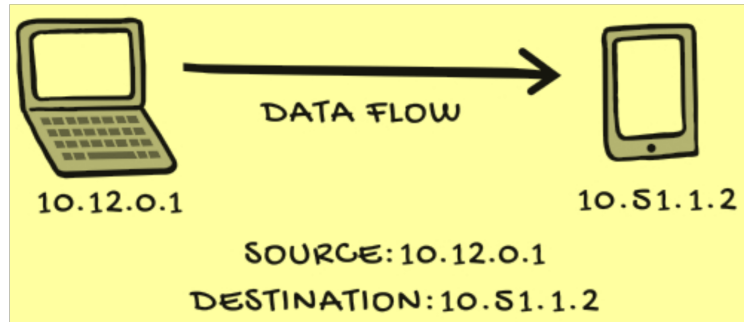
01111011.01011001.00101110.01001000

1 Byte=8 Bits

4 Bytes =32 Bits



Communication between a User & Web Server



IP Address Allocation

- Can be assigned in two ways:
 - Statically
 - Dynamically, by using the Dynamic Host Configuration Protocol (DHCP)
- Typically, servers are configured with **static IP addresses**, and end-user devices are configured with **dynamic IP addresses**.
- If the system is unable to find a DHCP server, it assigns itself an address using Automatic Private IP Addressing (APIPA).
 - APIPA addresses all begin with 169.254.
 - IP addresses like 169.254 indicate address allocation problems

Identifying valid IPv4 Addresses

- No octet in an IP address should ever be larger than 255.
- IP addresses starting with the number 127 are reserved for use as loopback addresses.
- The first number in an IP address should never be higher than 223.
 - Addresses starting in the range 224 – 239 are actually multicast addresses
 - Addresses starting in the range 240 – 255 are used for experimental purposes



Network Communication

Protocols

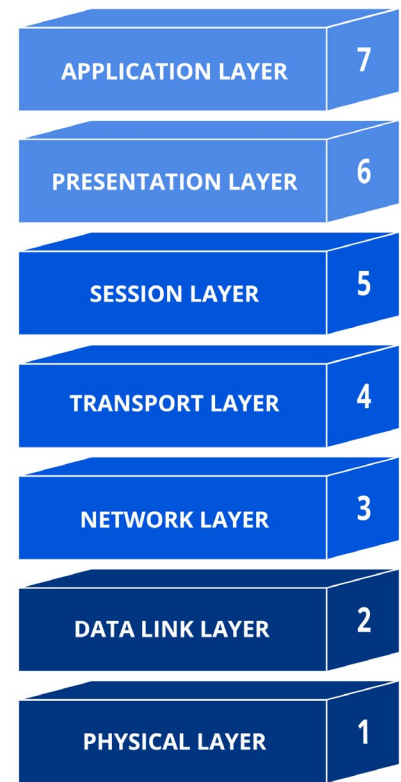


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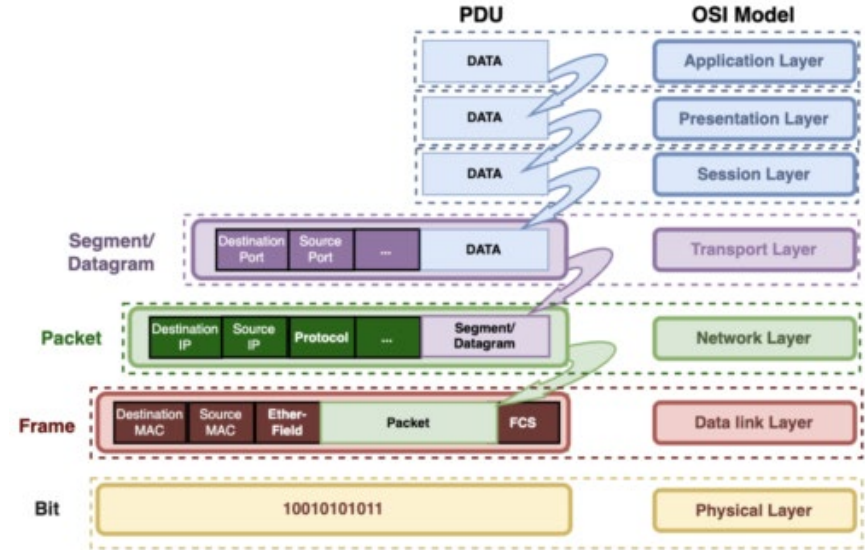
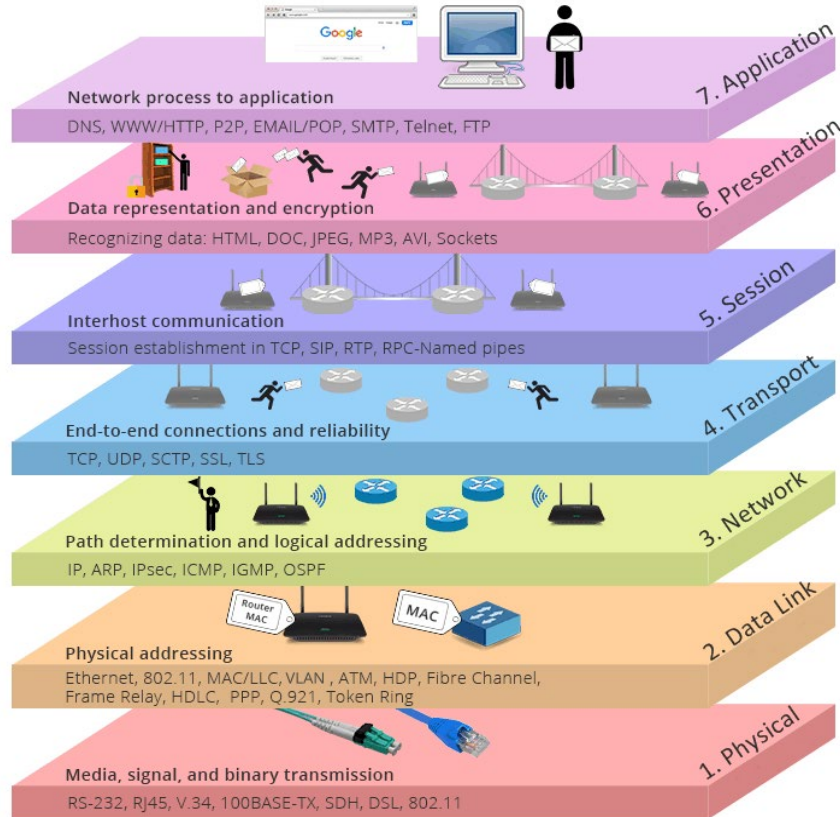
Open Systems Interconnection (OSI) Model

7. **Application layer:** determines how users interact with the data, using web browsers and other client applications. Protocols like HTTP, SMTP, and SSH exist at the Application layer.
6. **Presentation layer:** translates data so that it can be transmitted on a network. This layer determines how to represent a character in terms of bits and performs encryption and decryption
5. **Session layer:** manages the exchange of communications between systems
4. **Transport layer:** creates connections between systems and transfers data in a reliable manner. TCP and UDP are transport layer protocols.
3. **Network layer:** expands networks to many different nodes. The Internet Protocol works at this layer.
2. **Data-Link layer:** transfers data between two nodes connected to the same physical network.
1. **Physical layer:** sending bits over the network using wires, radio waves, fiber optics, and other means.



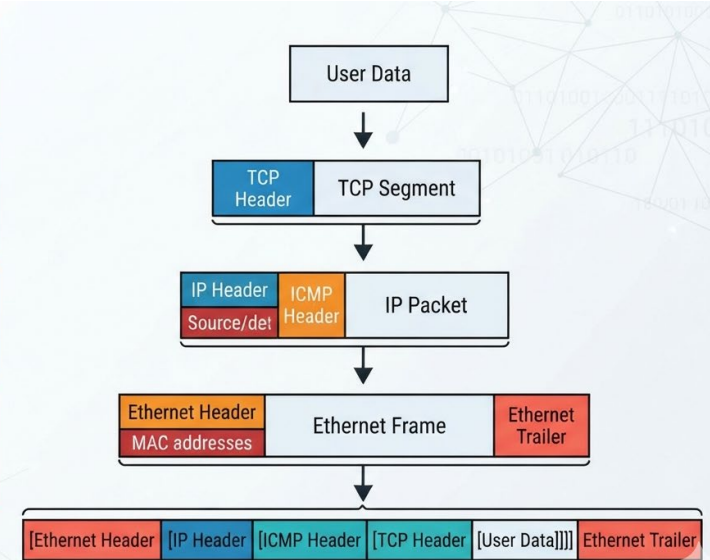
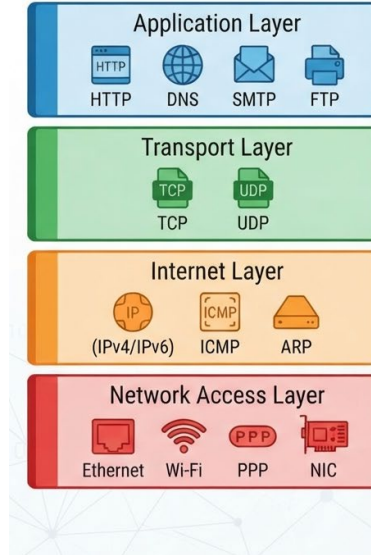
The Open Systems Interconnection (OSI) Model

أكاديمية كاوست
KAUST ACADEMY

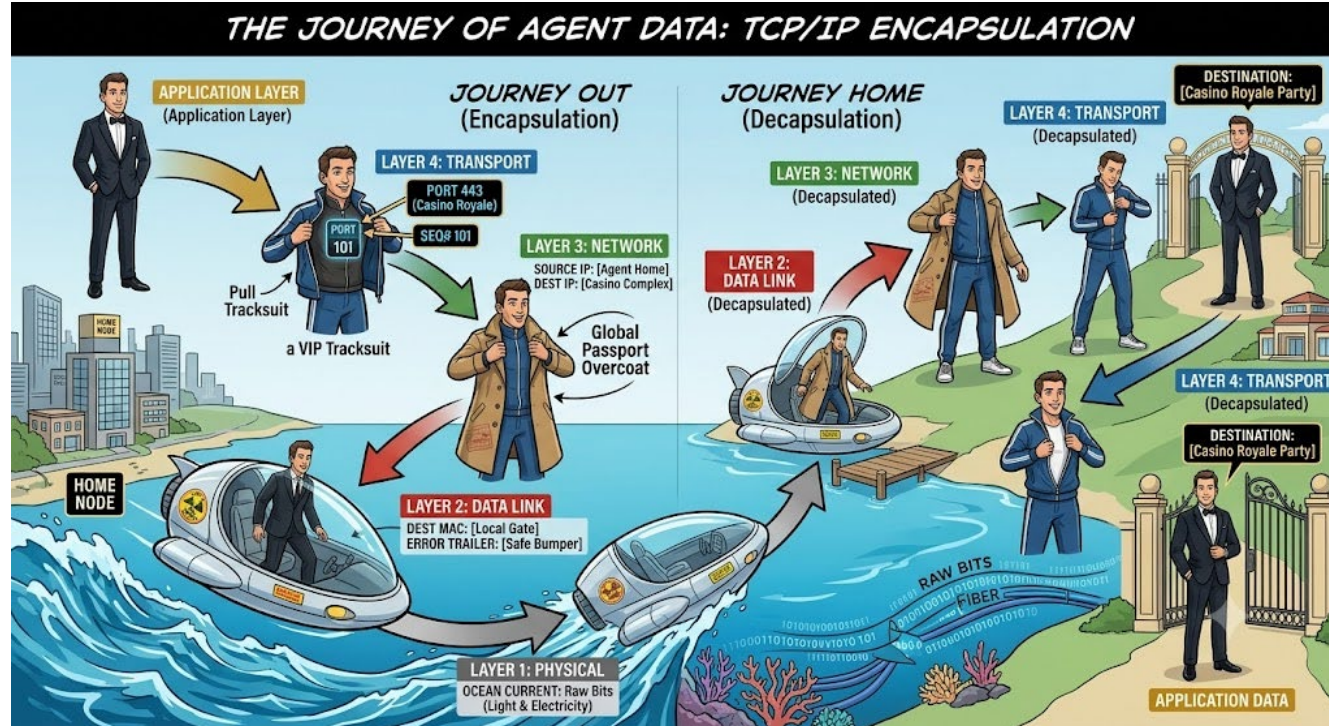


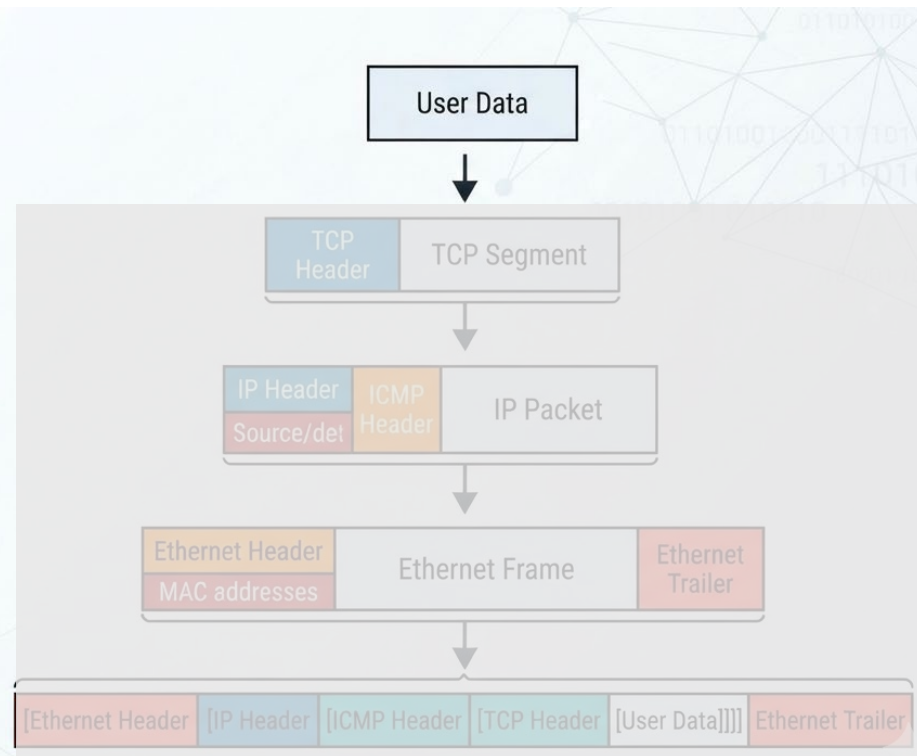
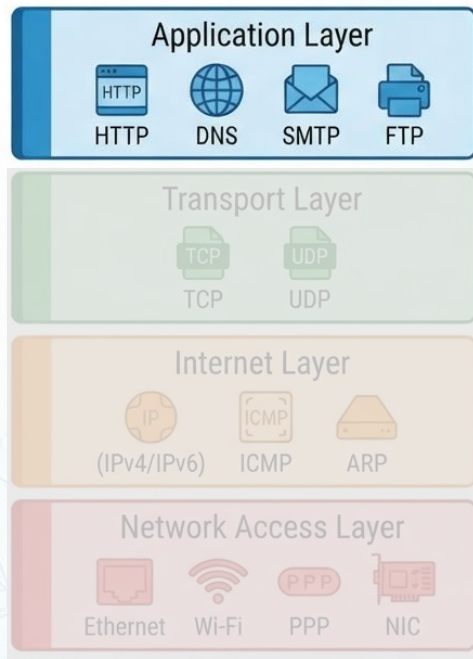
TCP/IP Networking

- Protocol that runs the Internet and basically every LAN on the planet
- The acronym TCP/IP has two parts.
 - Transmission Control Protocol (TCP)
 - Internet Protocol (IP)
- Each of these protocols plays a vital role in making sure that data gets from **Point A** to **Point B**.



Demystifying TCP/IP Protocol (analogy)





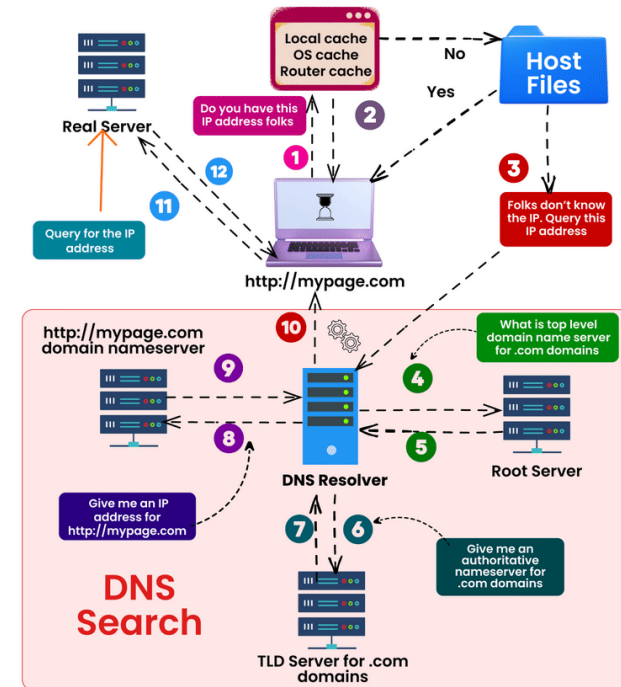
4a - Domain Name System (DNS)

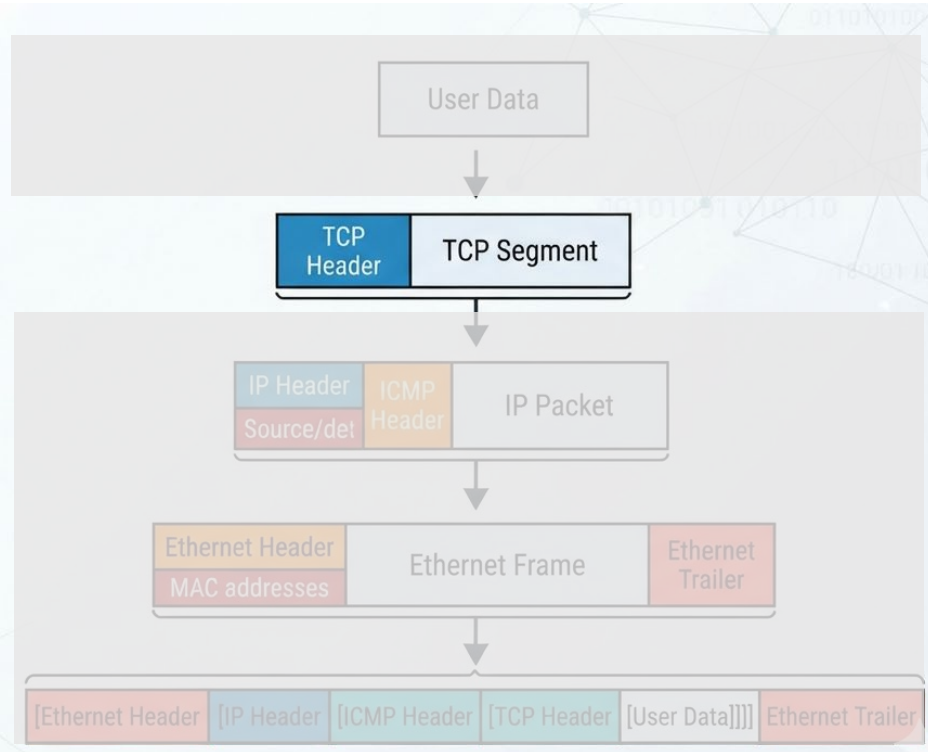
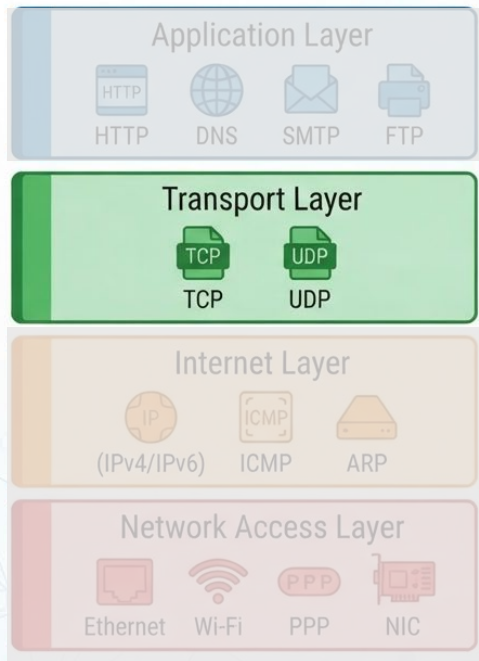
- IP addresses cannot be remembered by human beings
 - It is difficult to remember [109.171.129.32](#) as compared to [kaust.edu.sa](#)
- We want people to use easily recognizable names in place of IP addresses which is made possible by using DNS
- DNS is responsible for translating the uniform resource locator (URL) addresses that are commonly used for websites into the IP addresses associated with the servers supporting those sites.
- Terminology
 - Domain Name: top-level name that may be registered by a company, organization, or individual
 - URL: address of a specific web page or other resource hosted on that domain

How DNS Works?

- Every time you connect to a network, that network provides your computer with the **IP address of a local DNS server** that it can use to look up IP addresses
- Whenever you type in the domain name of a website in your browser, your computer sends a **request to the local DNS server** asking it for the IP address associated with that name
 - If the **server knows** the answer to your question, it simply responds to the request with the IP address, and then your web browser can go ahead and connect to the website using its IP address.
 - If the local DNS server doesn't know the answer to your question, it **contacts other DNS servers** to determine the correct answer and then responds to you
- Organizations who own domain names designate DNS servers that are the **responsible sources of information** about their domain name
- When a local DNS server needs to perform a lookup, it asks a series of questions that eventually lead it to the definitive answer from the DNS server that is responsible for a domain.

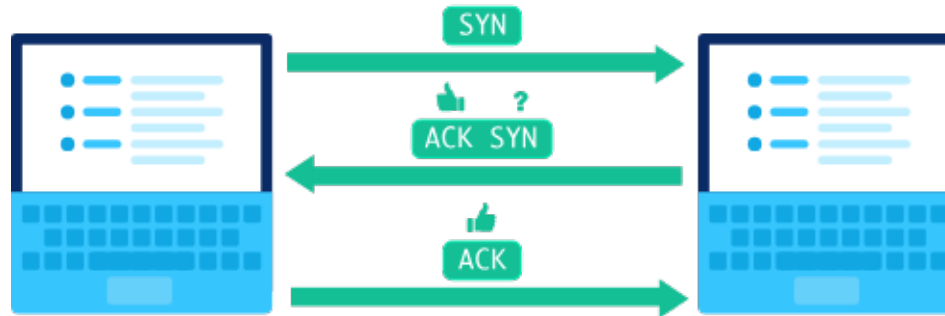
HOW DNS WORKS





3a - Transmission Control Protocol (TCP)

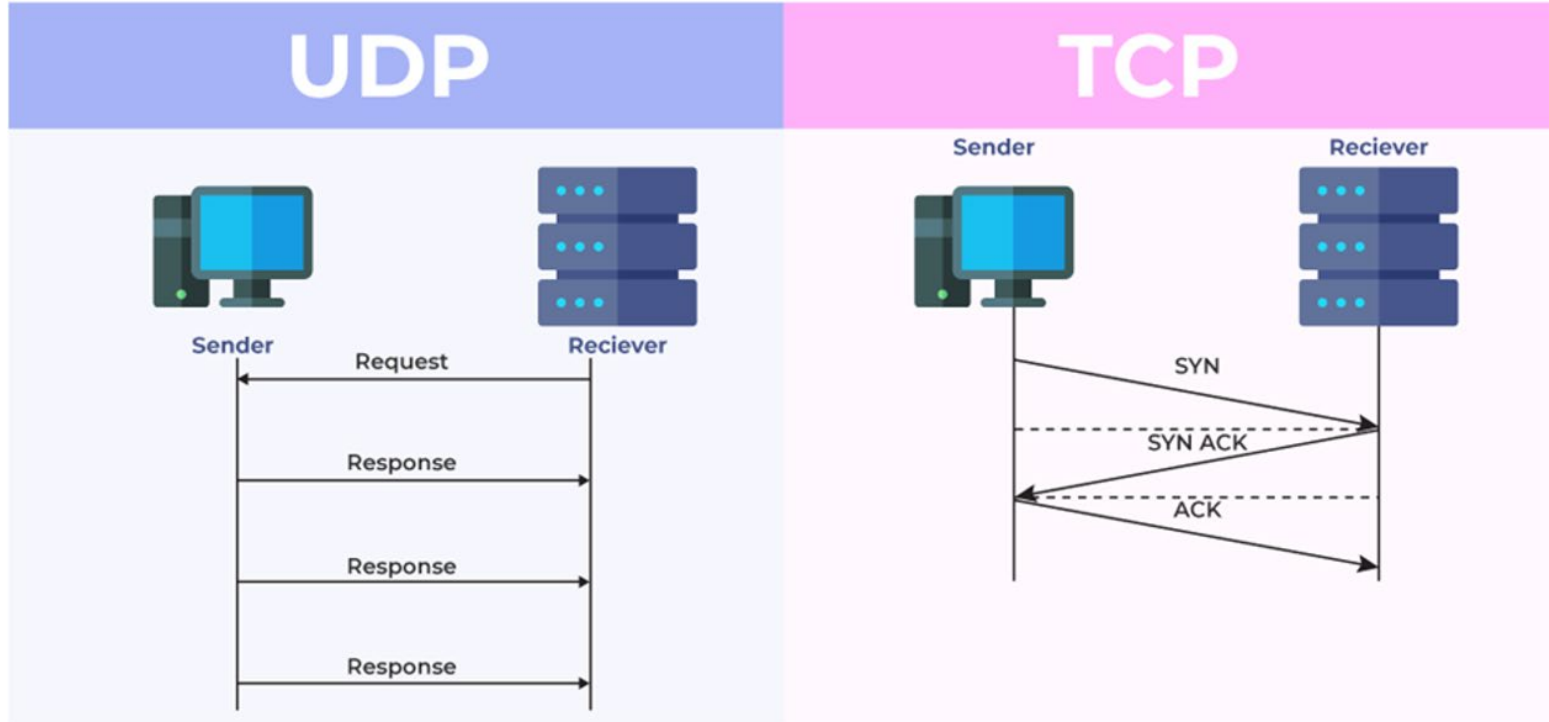
- TCP is responsible for **setting up** and **tearing down** connections between source and destination systems.
- TCP is also responsible for reliable delivery of packets
 - Tracking missed/lost/damaged packets
 - resend lost packet + correctly reordering
- Correctly reordering the packets at their destination, confirming a reliable delivery.



3b - User Datagram Protocol (UDP)

- Alternative to TCP
- It has **less overhead** and hence **faster** than TCP
- UDP not guarantee the delivery of packets from one system to another hence it is less **reliable** than TCP
- Used where speed is more important than ensuring every single packet delivery
 - Streaming video
 - VoIP telephony

TCP vs UDP Transmission



Network Ports

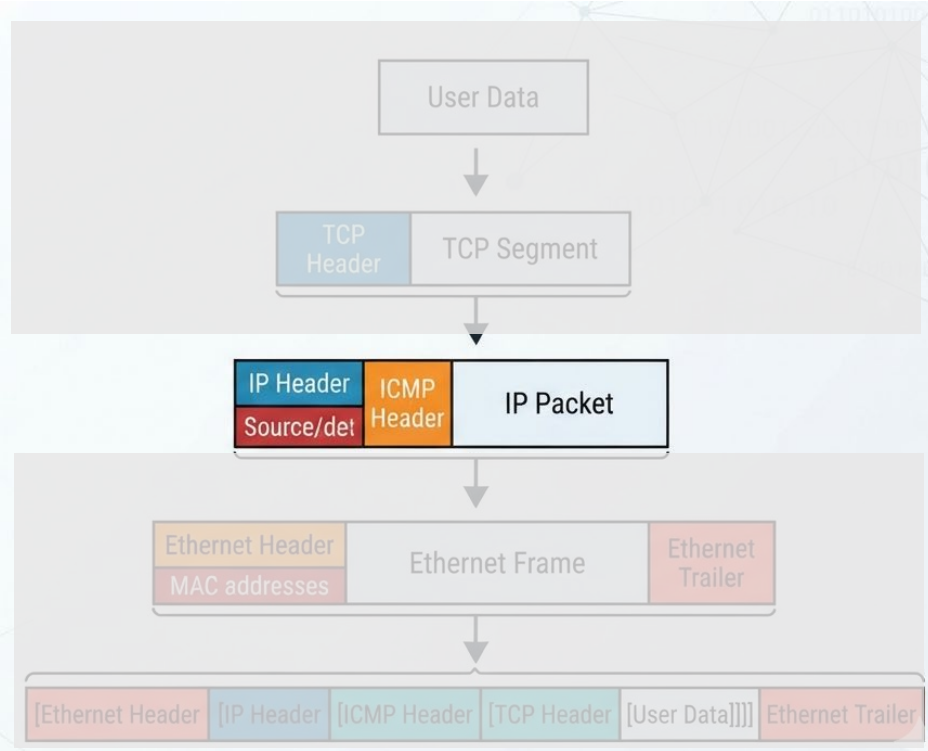
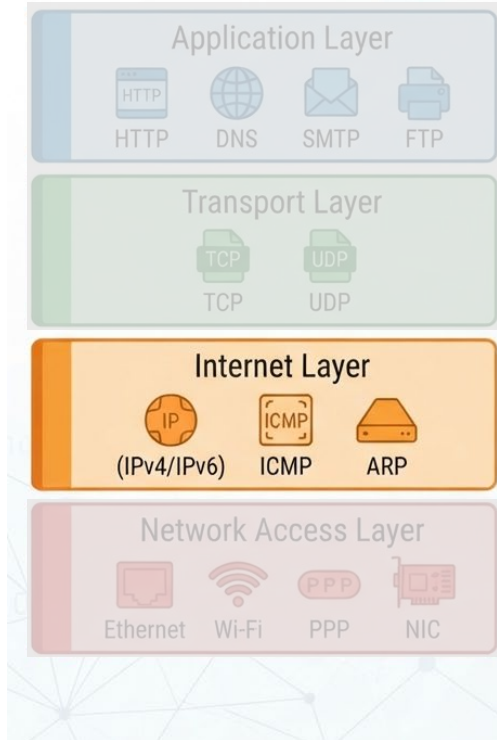
- Analogy
 - Street Address of a building -> IP Address (of a machine)
 - Apartment number -> Port Address (of an application)
- *Network ports* are particular locations on a system associated with a specific application.
- Network port are represented using a number ranging from 0 to 65,535.
 - Ports between 0 and 1,023 are called *well-known ports*.
 - Ports between 1,024 and 49,151 are known as *registered ports*
 - Microsoft reserved port 1433 for SQL Server database connections
 - Oracle registered port 1521 for its own database's use
 - Ports above 49,151 are set aside as *dynamic ports*.
 - applications can use on a temporary basis.

Well-Known Ports: Unencrypted vs Encrypted

Must-know commonly used ports to memorize

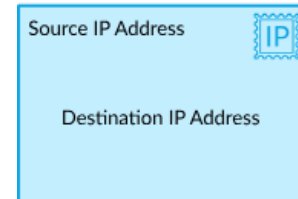
20-data 21-command control	FTP FTPS	data-989 control-990
25	SMTP SMTPS	465
80	HTTP HTTPS	443
110	POP3 POP3S	995
119	NNTP NNTPS	563
143	IMAP IMAPS	993
23	Telnet SSH/SCP /SFTP	22
53 123	DNS NTP	

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2a - Internet Protocol (IP)

- IP is responsible for routing information over the network (Internet, LAN)
- IP assigns each computer on the network its own address, called an IP address.
- IP addresses uniquely identify computers on a network and are the way that computers identify each other on an IP network
- When data travels on a TCP/IP network, the Internet Protocol breaks it up into small pieces called packets
- Packets are few kilobytes of data
- Large files are broken down to thousands of packets
 - To make data transfer reliable
 - To avoid network clogging

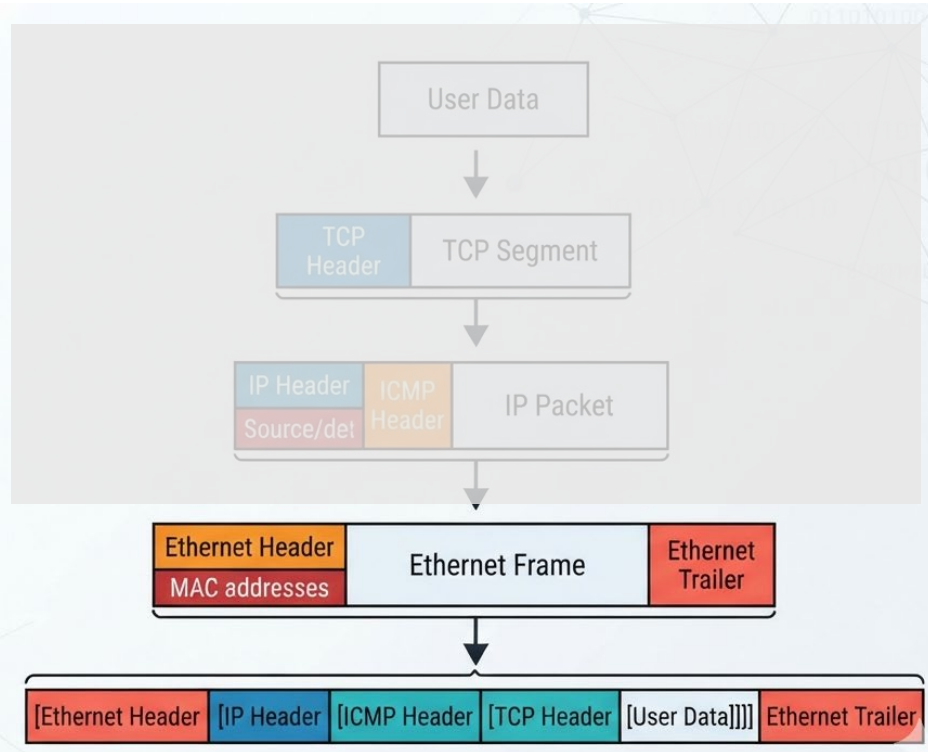
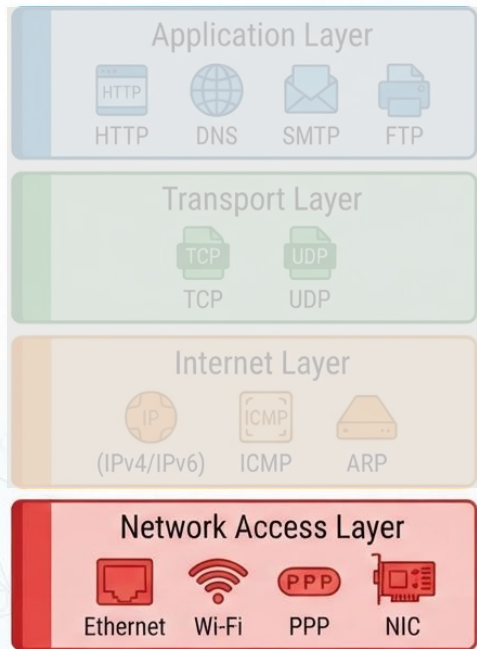


2b - Internet Control Message Protocol (ICMP)



- Housekeeping protocol of the Internet.
- Part of the TCP/IP Suite
- It's designed to allow networked devices and systems to communicate with each other about the operation of the network
- Uses
 - to detect whether remote systems are live on the network
 - to discover the network path between two systems
 - to report issues with network devices





Securing WI-FI Networks

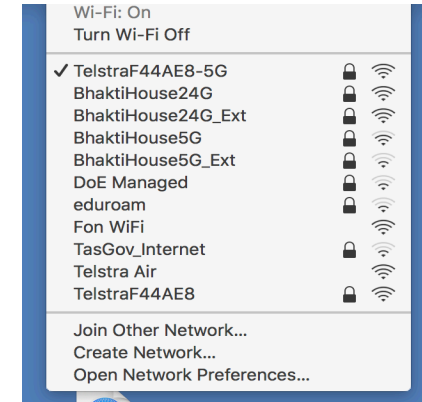
- ❖ *to protect from eavesdropping attacks*
- ❖ *to protect the network from unauthorized access*



Best Practices to Protect your Wireless Network



- Disable **Service Set Identifier** (SSID) broadcasting
 - No more announcing beacon frames of its presence, i.e. SSID not visible
 - It can still be found using simple hacking techniques (<https://www.youtube.com/watch?v=uGU-PDxpbbo>)
- Change Default Passwords
- Wireless Security
 - ❖ Open Networks – Not recommended **XXXXXX**
 - ❖ Preshared keys (PSK)
 - **Changing** is a tremendous burden due to need for **reconfiguring** all wireless devices
 - Prevents the identification of individual users and the **restriction of access by user identity** (hence, if a user leaves the organization, **key rotation** becomes necessary)
 - ❖ Enterprise Authentication
 - ❖ Captive Portals
 - ❖ Wireless Encryption



Wireless Encryption

Radio waves are vulnerable to Eavesdropping, hence we must **Encrypt** the traffic

- Wired equivalent privacy (WEP)
 - WEP was used for a long time and quick fix to achieve wired equivalent security
 - Serious design flaws -> not secure at all
- Wi-Fi protected access (WPA)
 - WEP successor (2003)
 - Temporal key integrity protocol (TKIP) changes the encryption key for each packet
 - Also found to be insecure
- Wi-Fi protected access v2 (WPA2)
 - WPA successor (2004)
 - AES Encryption - Counter mode cipher block chaining message authentication code protocol (CCMP)
 - Some potential issues but still being used
- Wi-Fi protected access v3 (WPA3)
 - New wireless devices since 2020 must support WPA3
 - Uses CCMP + Simultaneous authentication of equals (SAE) for initial communication setup

Wireless Encryption Summary

STANDARD	SECURITY STATUS
OPEN NETWORK	INSECURE
WEP	INSECURE
WPA	INSECURE
WPA2	SECURE
WPA3	SECURE



Network Protections

- **Secure Network Design (VPN, Segmentation, VLAN)**
- **Firewalls, IDS, IPS**
- **SEIM (Wazuh)**

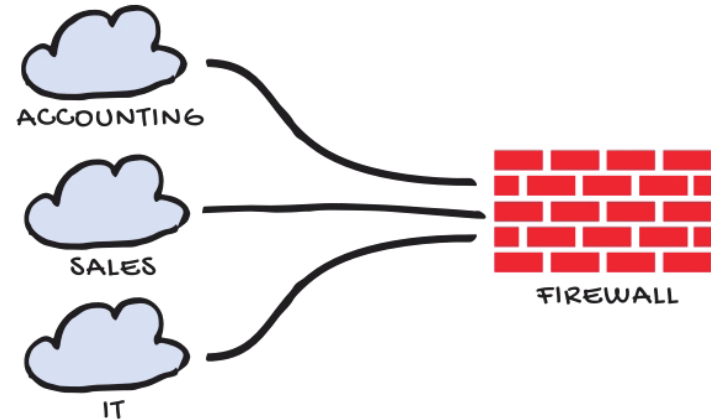
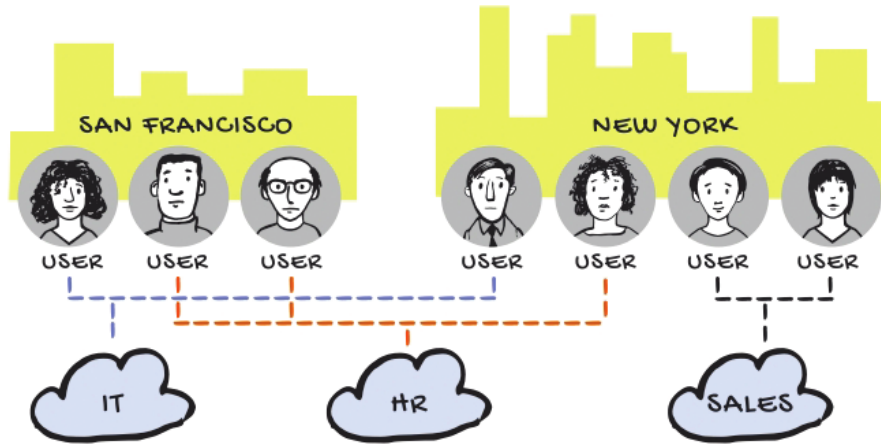


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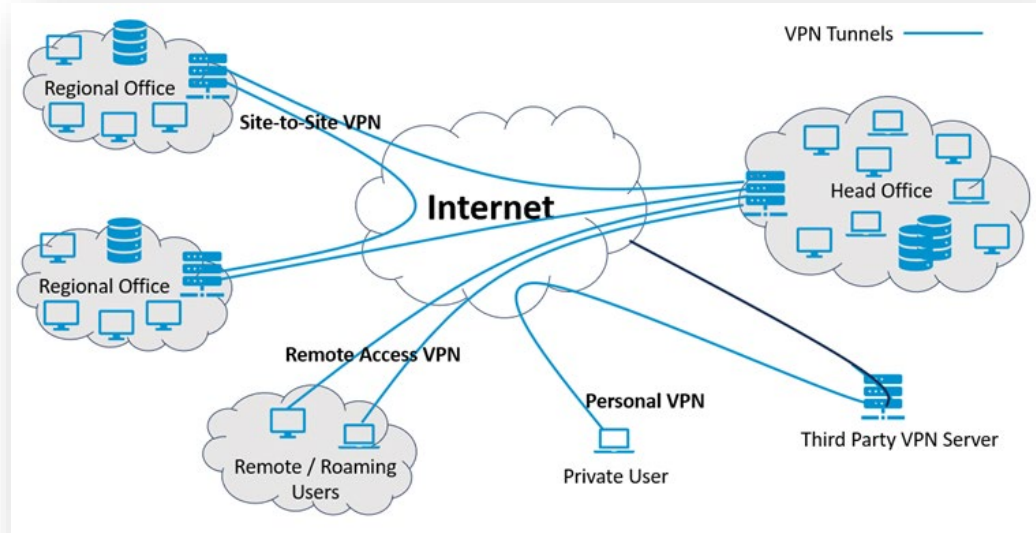
Virtual LANS (VLANs)

- An important network security control
- VLANs enable you to logically group together related systems, regardless of where they normally exist on the network.



Virtual Private Networks (VPNs)

- Provide two important network security functions to IT administrators:
 - Site-to-site VPNs
 - Remote access VPNs
- VPNs work by using **encryption** to create a virtual tunnel between two systems over the Internet.
- VPN endpoints
 - Firewall
 - Router
 - Server
 - Dedicated VPN concentrator



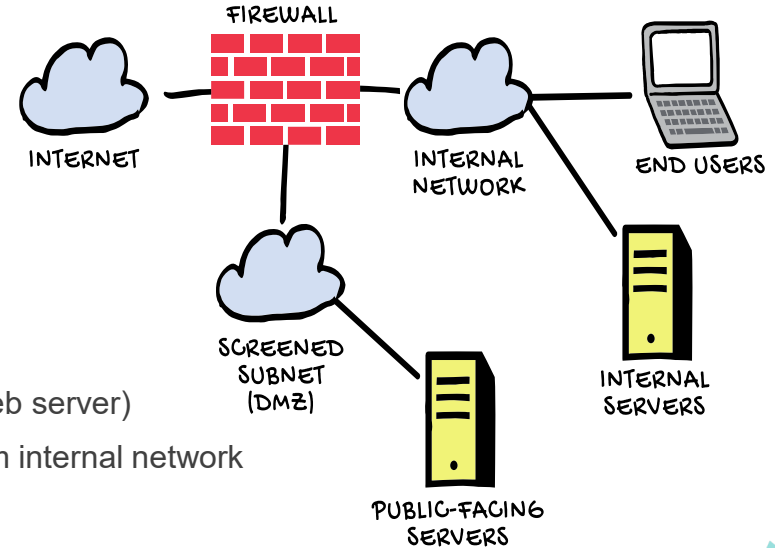
Firewalls

- Serves as the security guards of the network, analyzing all attempts to connect to systems on a network
- Determines whether the request should be **allowed** or **denied** according to the organization's security policy.
- Firewalls often sit at the network perimeter, in between an organization's routers and the Internet
- monitors inbound (ingress) and outbound (egress) traffic

Network Firewalls Divide Networks into Zones

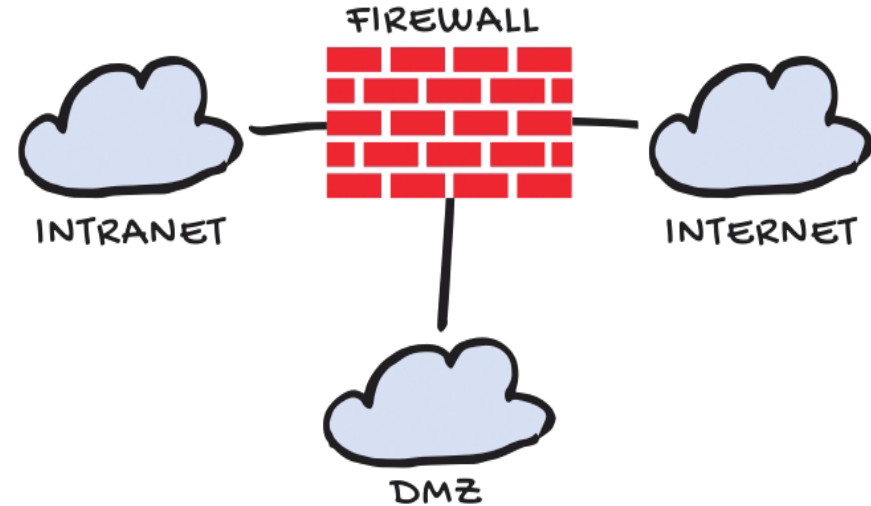
- Three network interfaces

1. Connects to the Internet (outside world)
 - Egress allowed
 - Ingress restricted
2. Connects to Intranet (local network)
3. Connects to Demilitarized Zone (DMZ)
 - Systems which must be open for public (email server, web server)
 - Vulnerable to attacks and compromises but separate from internal network



A Network Border Firewall with Three Interfaces

- Firewalls are security devices that sit on a network and restrict the communications that can pass through them
- Firewall groups systems into network segments called **security zones** based on each **system's security level**
- Typical border firewalls have **three network interfaces**



Endpoint Compliance

- Networks using this technology verify that devices connecting to the network comply with the organization's security policy before granting broader access
- Posture checks might include the following:
 - ✓ Verifying that antivirus software is running on the system
 - ✓ Validating that the antivirus software has current signatures
 - ✓ Ensuring that the device has a properly configured host firewall
 - ✓ Verifying that the device contains all recent security patches
- If posture check fails -> machine placed in a quarantined VLAN segment
 - Limited internet access
 - Allows to upgrade posture and join back
- Compliance check performed through in-device agents



INTRUSION DETECTION AND PREVENTION



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Intrusion Detection Systems (IDS) + Intrusion Prevention Systems

- **Intrusion Detection Systems** sit on the network and monitor traffic, searching for signs of potentially malicious traffic
- Examples
 - a request bound for a web server contains a SQL injection attack
 - a malformed packet is attempting to create a denial of service
 - a user's login attempt seems unusual based on the time of day and prior patterns
 - a system on the internal network is attempting to contact a botnet command and control server
- **Intrusion Prevention Systems** are just like intrusion detection systems but with a twist: *“They can take immediate corrective action in response to a detected threat.”*

- Environment Monitored
 - *Host-based IDS (HIDS) and host-based IPS (HIPS)* monitor and protect individual devices like computers and servers.
 - *Network-based IDS (NIDS) and network-based IPS (NIPS)* focus on safeguarding the entire network.
- Detection Errors
 - False Positive Errors: system alerts administrators to an attack but the attack does not actually exist
 - False Negative Errors: an attack actually takes place but the IDS does not notice it.

IDS/IPS Detection Techniques

■ Signature-based detection

- uses large databases that have patterns of data (or signatures) known to be associated with malicious activity
- When the system spots network traffic matching one of those signatures, it triggers an intrusion alert
- **Cannot detect a new intrusion**
- **Low false positive rate - reliable**

■ Anomaly-based detection

- develop a model of **normal activity** and then report **deviations** from that model as suspicious.
- **Potential to notice new attack types**
- **High false positive error rate**

Logging

- Treasure trove information for:
 - investigating an incident,
 - troubleshooting a technical problem,
 - gathering evidence.
- Objectives of logging
 1. **Accountability (identity attribution):** who caused the event (a specific person, a computer's IP address, or a geographic location)
 2. **Traceability:** track down all other events related to the investigated event
 3. **Auditability:** clear documentation of those actions

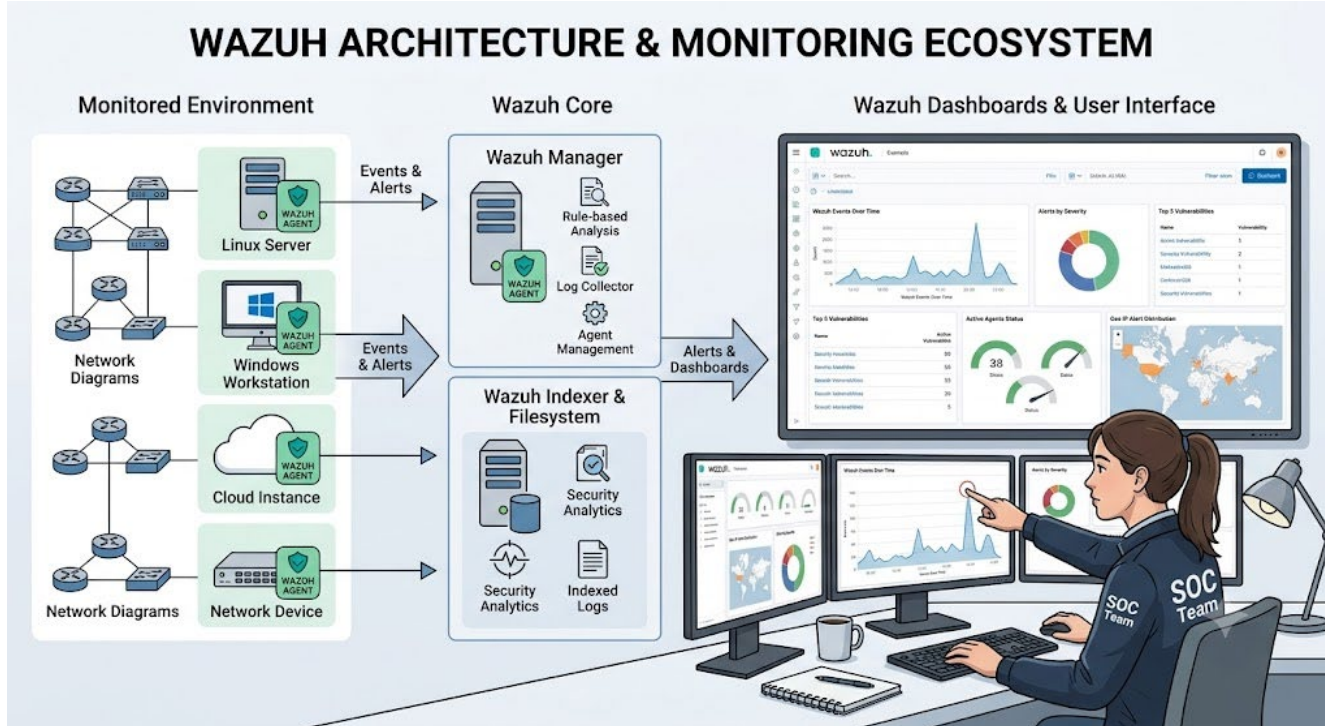
Log Monitoring

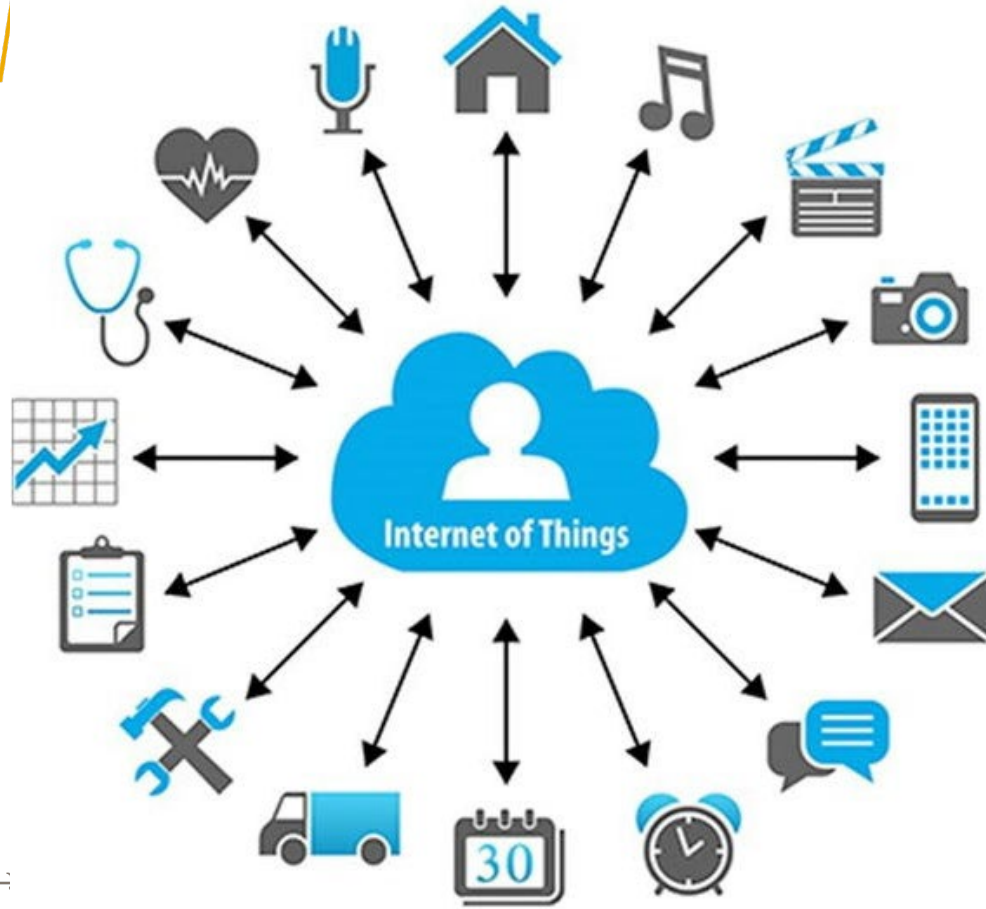
- Manual monitoring of logs is impossible (too many logs events)
- Software programs (using AI algorithms) perform security log analysis
- Security information and event management (SIEM) systems
 - act as a central, secure collection point for log entries
 - all of systems, network devices, and applications send log records directly to the SIEM
 - SIEM stores them in a secure fashion where they are safe from unauthorized modification and available for analysis.
 - Apply artificial intelligence to correlate all of those log entries to detect patterns of potentially malicious activity.

SIEM – Central Monitoring

- **Intrusion Detection System (IDS)** notices the unique signature of an attack in inbound network traffic (log event reported to SIEM)
- **Firewall** might note an inbound connection to a web server from an unfriendly country (log event reported to SIEM)
- **Web Server** finds suspicious queries that include signs of a SQL injection attack (log event reported to SIEM)
- **Database Server** finds a large query from a web application that deviates from normal patterns (log event reported to SIEM)
- **Router** reports a large flow of information from the database server to the Internet (log event reported to SIEM)

****In isolation, each of these activities may seem innocuous, but when the SIEM puts those pieces together, a pattern of suspicious activity emerges****





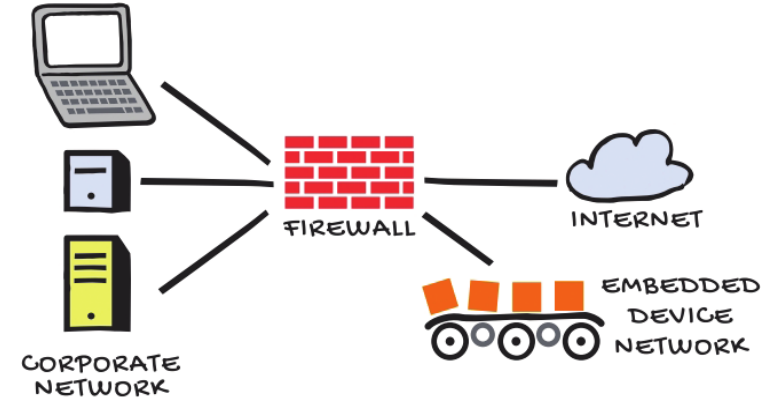
Internet of Things (IOT)

- Embedded computer systems are now part of our everyday life as we adapt to the *Internet of Things (IoT)*.
- Wireless smart devices everywhere (computer controlled + connected to Internet)
 - Smart Homes (smart garage, smart sprinkler)
 - Smart Cities
 - Smart Motorways, Smart Vehicles
 - Smart Industries (Industry 4.0)



IoT Security

- All of these IoT devices come with security challenges:
 - IoT devices usually use slimmed down versions of traditional operating systems (missing security extensions)
 - Difficult for the consumer to update the software
 - Connect to the same wireless networks that are used for personal productivity
 - Connect back to cloud services for command and control -> potential attack vector
- Network design solution: IoT network segmentation
 - A DMZ for IoT devices



IoT network segmentation



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